



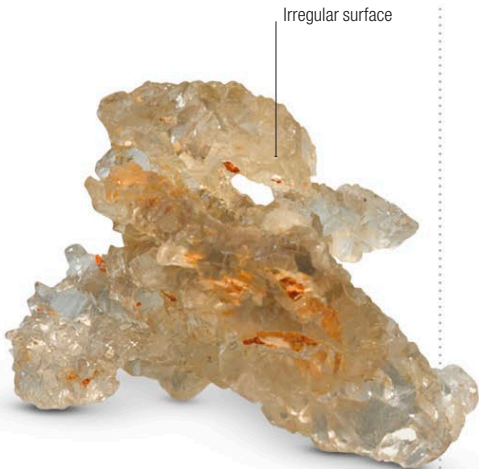
△ **Fine**, mixed-cut cushion petalite

Petalite

Petalite's name comes from the Greek word for "leaf", a reference to its tendency to break into thin, leaf-like layers. It is usually found as masses of small crystals and, rarely, as individual crystals. In its massive form it is cut *en cabochon*, and its colourless, transparent crystals are faceted for collectors only. Because it is brittle and easily split, it requires extreme care in the faceting process, and it is too fragile to wear. Facet-grade petalite is found principally in Brazil, which produces collectors' gemstones of up to 50 carats in weight.

Specification

Chemical name Lithium aluminium silicate | **Formula** $\text{LiAlSi}_4\text{O}_{10}$
Colours Colourless to greyish-white, pink, green | **Structure** Monoclinic | **Hardness** 6–6.5 | **SG** 2.4 | **RI** 1.50–1.52
Lustre Vitreous | **Streak** White | **Locations** Brazil, Sweden, Italy, Russia, Australia, Zimbabwe, Canada



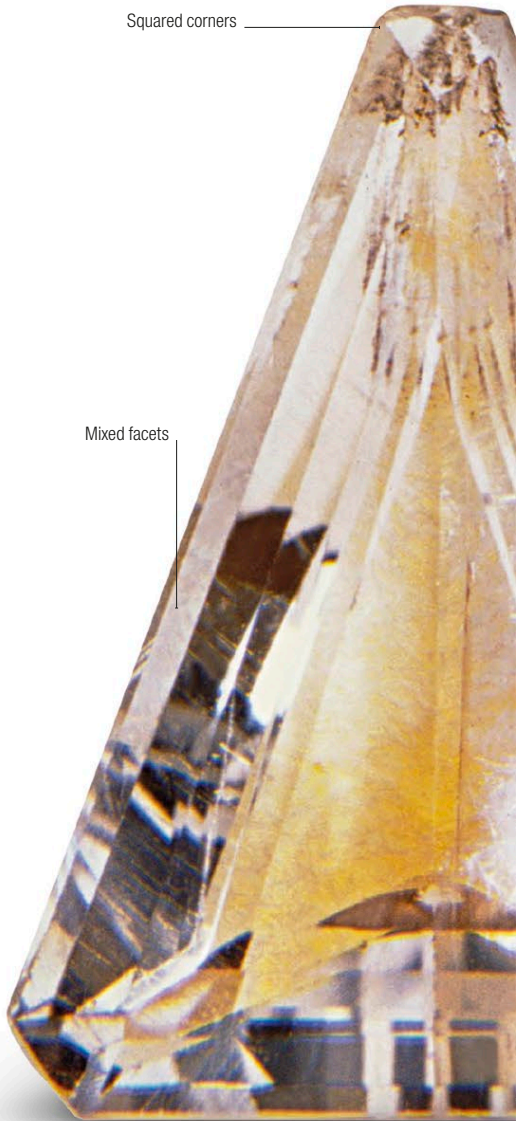
Unusual form | Rough | This striking grouping consists of several petalite crystals, naturally acid-etched, but retaining some gem material.



Cabochon grade | Rough | This rough specimen will be sliced and polished into cabochons, as the raw material needs to be of higher quality for faceting.



Polished petalite | Carved | In an unusual decision, a lapidary has shaped this petalite specimen into a baroque form, and then polished it to a smooth finish.

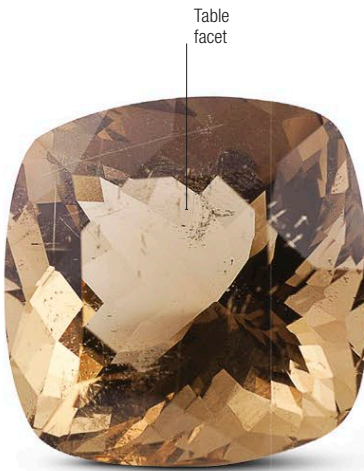


Mixed facets



Colour change | Colour variety | This rare 6.22-carat stone from Myanmar changes from olive-green in sunlight to fiery red in incandescent light.

The chemical element lithium was first discovered in petalite, which is still an important ore of the element



Burmese petalite | Cut | This flawless 25.20-carat smoky-brown, mixed-cut cushion petalite, also from Myanmar, is part of the Smithsonian Institution's gem collection.

Mixed cut | Cut | The corners of this yellow, triangular, mixed-cut petalite have been squared off to prevent breakage of the brittle material.